

NIDHI BERDE

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PROFESSIONAL SUMMARY

Software Engineer, 2+ years, backend and agentic AI systems in Python. Redesigned AI-model routing for a production document platform, cutting API costs 60% while maintaining 90% matching accuracy, verified by comparing API call volume before and after the change. Migrated the same platform to an event-driven AWS SQS/Lambda architecture, deployed via Terraform, improving server utilization and processing speed. Ships production REST APIs (FastAPI) and async Python pipelines, and works directly with business and compliance stakeholders to turn ambiguous requirements into deployed systems.

TECHNICAL SKILLS

Languages & APIs: Python, SQL (query authoring, optimization, troubleshooting), JavaScript, REST API development (FastAPI)

Data Pipelines & Streaming: Apache Spark / PySpark, Apache Kafka, Apache Airflow, Informatica PowerCenter, Hive / Hadoop ecosystem

Data Formats: JSON, XML, Parquet (schema evolution & compatibility)

Cloud & Storage: AWS (S3, EC2, SQS, Lambda) - Certified, GCP BigQuery, Azure OpenAI, MongoDB/DocumentDB, PostgreSQL, MySQL, SQL Server

DevOps & Quality: CI/CD, Git, Azure DevOps, Docker, Terraform, data validation & root-cause analysis

AI/ML & Agentic Systems: LangChain, LangGraph, GPT-4 Vision, TF-IDF, Claude Code, Ollama, prompt engineering

PROFESSIONAL EXPERIENCE

Software Engineer - Data Engineering & AI Platform | Tata Consultancy Services, Edison, NJ

June 2024 - Present

- Built and maintained the REST APIs behind an AI-powered document compliance platform for an aerospace & defense client, automating the discovery, classification, and marking of **10,000+ regulated documents across 30+ file formats**.
- Designed a fast, low-cost way to match documents to their source: text-pattern matching (TF-IDF) resolves most cases instantly, with an AI vision model stepping in only for uncertain matches, **cutting API costs by 60% while maintaining 90% matching accuracy**.
- Built an LLM-powered data-quality framework that uses a table's schema and sample records to generate validation checks automatically and answer natural-language questions over the data, **removing manual test-query writing and giving non-technical stakeholders self-service access to the data**.
- Redesigned the platform's database structure and migrated the document-processing pipeline from a custom worker-pool system to an event-driven architecture that reads from the updated database, where each request queues in AWS SQS and triggers a Lambda to run the file-scanning pipeline, **improving server utilization and processing speed, with infrastructure defined and deployed via Terraform**.
- Delivered KPI/metrics dashboards and on-demand report exports (Excel, JSON) that gave business and compliance stakeholders self-service visibility into platform activity, removing the need for manual reporting.

Data Engineering Intern | Tata Consultancy Services, Edison, NJ

June 2023 - August 2023

- Built end-to-end ETL pipelines using Informatica PowerCenter on a recurring schedule, integrating Oracle, SQL Server, and Excel sources into cloud data warehouses, applying validation, cleansing, and transformation logic to ensure data integrity and consistency across distributed systems.
- Developed a Python/Tkinter desktop tool that documents legacy ETL jobs automatically by parsing configuration exports up to 200MB each and generating an inventory spreadsheet, removing a manual, error-prone step from the pipeline documentation process.

Application Development Associate | Accenture, Mumbai, India

January 2021 - August 2022

- Executed migration of 30+ TB from on-premises Hadoop (Hive/HiveQL) to GCP BigQuery for a global FMCG client, orchestrating the pipeline with Apache Airflow DAGs, converting HiveQL to BigQuery SQL, and applying partitioning strategies that **improved query performance by 35%**.
- Designed and maintained 15+ curated BigQuery data warehouse tables spanning raw, refined, and aggregated layers, conducting reconciliation and integrity checks **across 30+ supply chain datasets post-migration**.

PROJECTS

Newsletter Intelligence Pipeline | Python, Claude API, Ollama - Personal Project

- Architected a 5-stage async pipeline that connects to Gmail via OAuth and turns 20+ newsletter subscriptions across 4 categories into a ranked daily brief, using an LLM-agnostic client (Claude API or local Ollama) with a hybrid heuristic + LLM classification step that cuts unnecessary LLM calls before falling back to model-based classification, and a deduplication layer that merges overlapping stories across sources instead of naive concatenation.

Product Ingredient Plausibility Agent | Python, LangChain, LangGraph, Claude API, Ollama - Personal Project

- Built a two-agent system (LangChain, LangGraph) where a Researcher agent extracts a product's ingredients and marketing claims from the web via a Tavily search tool, and a Biologist agent cross-references each ingredient's real molecular properties against the PubChem database to score scientific plausibility using pharmacological rules (Lipinski's 500 Da threshold, logP, TPSA), with a conditional retry loop that routes low-confidence verdicts back to the Researcher for more evidence before finalizing - runs on either the Claude API or a local Ollama model, with cached results for repeat lookups.

EDUCATION

M.S. Computer Science (2024) | Pace University, New York, USA | GPA: 3.8

B.E. Computer Engineering (2020) | University of Mumbai, India

CERTIFICATIONS

AWS Certified Cloud Practitioner | AWS Certified Solutions Architect - Associate